

Probability & *statistics for AI.*

From observations to evidence.

Learn what the numbers do-and do not-tell you.

🕒 22-27 min core reading · Build on Chapter 6 · 12 short sections

An AI result is evidence, not certainty. Probability describes uncertainty; statistics helps you learn from observations. The useful habit is to ask what was measured, what was assumed, and what the number actually means.

WHAT YOU WILL BE ABLE TO DO

Summarize data, distinguish two directions of conditional probability, calculate a Bayes update, and interpret a confidence interval. Recognize why correlation and a small p-value cannot establish a complete conclusion.

Reading path: Follow a synthetic job-monitoring example, change its alert rates, then reproduce the calculations in Python. Optional notes add mathematical depth. Every dataset and rate in this chapter is invented for learning.

1. Start with the population and the sample

A **population** is the full group you want to describe: for example, all jobs your service will process next month. A **sample** is the subset you observe. A **parameter** describes the population, such as its true mean runtime; a **statistic** is calculated from a sample to estimate or describe something, such as the sample mean.

Descriptive statistics summarizes the observed data. **Statistical inference** uses data and assumptions to draw conclusions about a wider population. Five observed runtimes tell you exactly about those five jobs; using them to predict next month's performance needs additional justification.

Collection matters before calculation. Logs containing only successful daytime jobs omit failures and overnight workloads: **selection bias**. Ten copies of one job are not ten independent observations. More rows do not automatically mean more independent information, and a larger biased sample does not repair the collection process.

For an AI evaluation, name the target population, sampling method, time period, and unit of observation. A random sample gives each eligible unit a specified chance of selection. It helps control selection bias, but missing records and a changing future population can still limit your conclusion.

2. Describe the center and the spread

Use five synthetic runtimes, in seconds: **[2, 3, 3, 4, 8]**.

Summary	Calculation	Meaning
Mean	$(2 + 3 + 3 + 4 + 8) / 5 = 4$	Arithmetic average
Median	Middle sorted value = 3	Middle position; average the two middle values for even counts
Mode	3 , occurring twice	Most frequent value; there may be ties
Range	$8 - 2 = 6$	Maximum minus minimum

The mean uses every value's magnitude. The median uses their sorted positions. Replacing 8 with 18 raises the mean to 6, while the median remains 3. Neither summary is universally best: choose the question, then inspect the distribution.

➤ The average and the middle need not agree

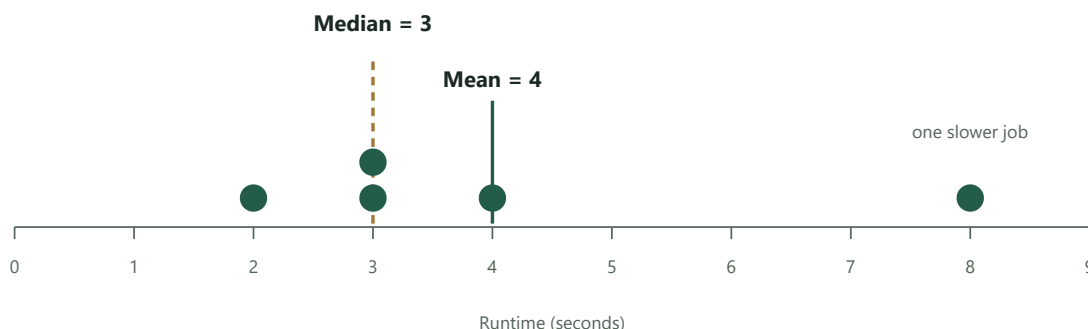


Figure 1. Each circle is one runtime observation. The two stacked circles at 3 seconds are separate jobs. Mean = 4 seconds; median = 3 seconds.

Variance measures squared deviations from a mean. The deviations here are $[-2, -1, -1, 0, 4]$; their squares sum to 22. The usual **sample variance** uses $n - 1$ in the denominator:

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1} = \frac{22}{4} = 5.5 \text{ s}^2, \quad s = \sqrt{5.5} \approx 2.345 \text{ s}$$

Here $n = 5$, x_i is one observation, $\bar{x}$ is the sample mean, and s is the **sample standard deviation (SD)**. Squaring prevents positive and negative deviations from cancelling; taking the root restores seconds. If these five values are the entire population being described, divide by 5 instead: population variance = 4.4 s^2 . See NIST's measures of scale.

▼ Go deeper: $n - 1$, percentiles, and standardization

After estimating the mean, the deviations must sum to zero; knowing $n - 1$ fixes the last one. For independent, identically distributed observations with finite variance, dividing by $n - 1$ makes the sample variance an unbiased estimator of population variance. This does not make every estimate accurate, nor make the square-root estimate of SD exactly unbiased.

A **quantile** marks a position in a distribution; a percentile uses a 0-100 scale. The median is the 50th percentile. The interquartile range is $Q_3 - Q_1$, the width between the 75th and 25th percentiles. Software uses different interpolation conventions for small datasets; record the convention. Five values cannot support a stable estimate of a population's extreme tail, such as p95 latency.

A **z-score** expresses a value relative to a chosen mean and SD: $z = (x - \text{mean}) / \text{SD}$. Using our sample statistics, 8 seconds becomes $(8 - 4) / \sqrt{5.5} \approx 1.706$. This feature standardization differs from dividing a whole vector by its length in Chapter 6. Fit preprocessing statistics on training data and reuse them on validation and test data. Constant features have SD zero and require explicit handling.

3. Read probability as a statement about an event

An **outcome** is one possible result; the **sample space** contains the possible outcomes. An **event** is a set of outcomes. Let F mean “the job fails” and A mean “the monitor raises an alert.” Probabilities lie between 0 and 1; multiplying by 100 expresses a percentage.

Expression	Read it as
$P(F)$	Probability of failure
$P(\text{not } F) = 1 - P(F)$	Probability of no failure
$P(F \cap A)$	Probability of both failure and alert
$P(F \cup A)$	Probability of failure or alert, including both
$P(F A)$	Probability of failure among alerted jobs

For “or,” add the two probabilities and subtract their overlap: $P(F \cup A) = P(F) + P(A) - P(F \cap A)$. Otherwise, the overlap is counted twice.

Conditional probability restricts the denominator to the condition:

$$P(F | A) = \frac{P(F \cap A)}{P(A)}, \quad P(A) > 0$$

$P(F | A)$ and $P(A | F)$ ask different questions. The first starts with alerted jobs; the second starts with failed jobs. Write “among which group?” before calculating.

▼ Go deeper: joint, marginal, independent, and mutually exclusive

$P(F \cap A)$ is a **joint probability**. $P(F)$ is a **marginal probability**, obtained by adding over the possibilities for the other event. Conditional probabilities hold one event fixed. A two-way count table makes all three visible.

F and A are **independent** if $P(F \cap A) = P(F)P(A)$. When $P(A) > 0$, this is equivalent to $P(F | A) = P(F)$: learning about A does not change F’s probability. This statistical independence differs from Chapter 6’s linear independence of vectors.

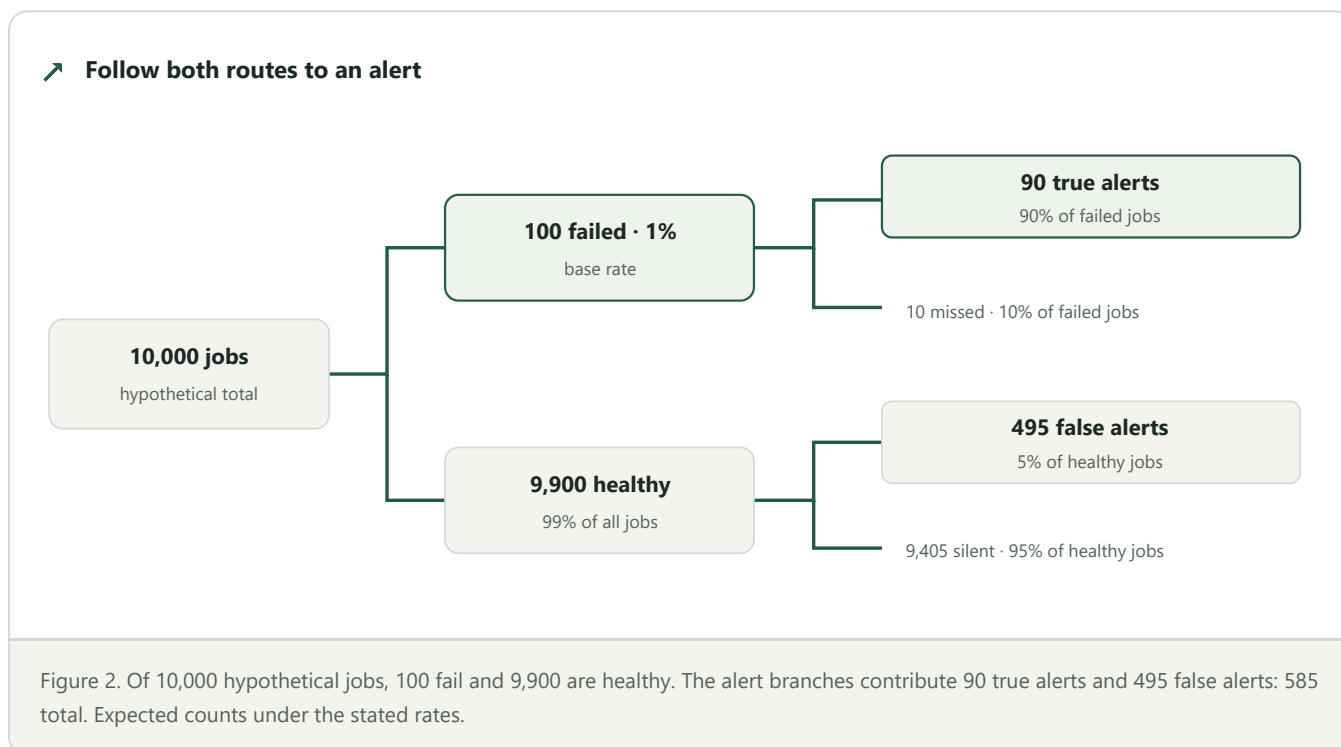
Mutually exclusive events cannot occur together. Two events with positive probability that are mutually exclusive cannot be independent: knowing one occurred rules out the other. Independence is an assumption to justify, not a synonym for “different things.”

4. Use Bayes rule to reverse the question

Suppose 1% of jobs fail. The monitor catches 90% of failures but also alerts on 5% of healthy jobs. For a hypothetical 10,000 jobs, the expected counts are:

Actual outcome	Alert	No alert	Total
Failed	90 true positives	10 false negatives	100
Healthy	495 false positives	9,405 true negatives	9,900
Total	585	9,415	10,000

These names treat failure as the **positive class**. Positive means the event being detected, not something desirable.



Of 585 alerts, only 90 correspond to failures: **90 / 585 ≈ 15.38%**. The 90% catch rate answers $P(A | F)$, not $P(F | A)$. Healthy jobs vastly outnumber failed jobs, so even a modest false-positive rate creates many false alerts.

Bayes' rule writes the same calculation in probabilities:

$$P(F | A) = \frac{P(A | F)P(F)}{P(A)} = \frac{0.90 \times 0.01}{0.90 \times 0.01 + 0.05 \times 0.99} \approx 0.1538$$

$P(F)$ is the **prior**, before seeing the alert. $P(A | F)$ is the probability of that evidence under failure, a **likelihood** term. $P(F | A)$ is the **posterior**, after observing the alert. The denominator counts every way to get an alert, including healthy jobs. No machine-learning training is needed to perform this update.

5. Explore how the base rate changes an alert

The **base rate** is the population frequency of the condition. In this example, **recall** = $TP / (TP + FN)$, **false-positive rate** = $FP / (FP + TN)$, and **precision** = $TP / (TP + FP)$. TP, FN, FP, and TN are the four interior table cells above. Precision is our $P(\text{failure} | \text{alert})$. These denominators are also used in Google's classification guide.

How much should you trust an alert?

Expected counts for 10,000 hypothetical jobs. Change the assumed rates; no random sample is drawn and no classifier is trained.

Compare the same 10,000 hypothetical jobs under four assumptions. FPR means false-positive rate.

Base / recall / FPR	True alerts	False alerts	P(failure alert)
1% / 90% / 5%	90	495	15.38%
10% / 90% / 5%	900	450	66.67%
1% / 90% / 0%	90	0	100%
1% / 0% / 0%	0	0	Undefined

Precision divides true alerts by all alerts. A zero denominator is undefined, not 0%. These are expected counts under stated rates, not a simulation or measured performance.

Change the failure rate from 1% to 10%, keeping recall at 90% and false-positive rate at 5%. Precision becomes **66.67%**, even though those two monitor rates stayed fixed. A test result measured on one population may not transfer unchanged to another.

The controls vary assumptions independently. In a real classifier, moving a decision threshold usually changes recall and false-positive rate together; population changes can change both too. The calculator is a mathematical thought experiment, not measured monitor performance.

▼ Go deeper: accuracy, specificity, and calibration

At the default settings, accuracy = $(TP + TN) / \text{total} = 94.95\%$. A monitor that never alerts would be 99% accurate because healthy jobs dominate, yet catch no failures. Accuracy alone can hide the behavior you care about.

Specificity is $TN / (TN + FP)$, or $1 - \text{false-positive rate}$. A false-positive rate is not the fraction of alerts that are false; that latter fraction is $1 - \text{precision}$ when there are alerts.

A probability predictor is **calibrated** when, across sufficiently many comparable predictions, outcomes agree with its stated probabilities: among predictions near 0.8, roughly 80% should be positive. Calibration is assessed over groups, not established by one outcome. It differs from accuracy and does not guarantee useful separation between classes; a constant base-rate prediction can be calibrated but uninformative about individual jobs.

6. Understand distributions and expected values

A **random variable** assigns a numerical value to an outcome. Y might be 1 for a failed job and 0 otherwise; T might be its runtime. A **distribution** describes which values are possible and how probability is assigned to them.

Distribution	Example	Assumptions to notice
Bernoulli	One failure indicator, 0 or 1	$P(Y = 1) = p$
Binomial	Number of failures in n trials	Independent Bernoulli trials with the same p
Normal (Gaussian)	A bell-shaped model for a continuous measurement	Described by mean μ and variance σ^2 ; real data need not follow it

For discrete values, a **probability mass function (PMF)** assigns probability to each value; those probabilities sum to 1. A continuous **probability density function (PDF)** assigns density. Probability is the area over an interval, not the height at one point. Density can exceed 1, and a variable with a continuous density has probability zero of equaling any one exact value.

An **expected value** is a probability-weighted average. In a separate toy runtime distribution, T takes 2, 4, or 10 seconds with probabilities 0.5, 0.3, and 0.2:

$$E[T] = 2(0.5) + 4(0.3) + 10(0.2) = 4.2 \text{ seconds}$$

No job in this model lasts 4.2 seconds. An expectation is not necessarily a possible outcome, the most likely outcome, or the next outcome. Population variance is the expected squared deviation from the mean: $\text{Var}(T) = E[(T - E[T])^2]$.

▼ Go deeper: cumulative probability, tails, and likelihood

A **cumulative distribution function (CDF)** gives $P(T \leq t)$. Unlike a density, it always lies between 0 and 1 and never decreases. A histogram estimates the distribution from observed values; its appearance depends on the bin choices.

Runtime data can have a long right tail. A normal distribution permits negative values, so it is not automatically a suitable model for positive runtimes. Examine skew, tails, and domain constraints. Investigate unusually large observations before deciding they are errors; a genuine slow job may be operationally important.

In **maximum likelihood estimation (MLE)**, hold the observations fixed and choose parameters making them most likely. Three failures in 100 independent trials with one shared failure probability give the Bernoulli MLE $p = 3/100$.

Likelihood is a function of parameters for fixed data; it is not, by itself, a probability distribution over those parameters. **Maximum a posteriori (MAP)** instead chooses a parameter maximizing its posterior, combining likelihood with a prior. Bayes' rule is the connection; optimization is the next chapter's topic.

7. Separate data spread from estimate uncertainty

SD describes variability of individual observations. **Standard error (SE)** describes variability of an estimator across repeated samples. For the mean of n independent observations from the same population with finite SD σ , $\text{SE} = \sigma / \sqrt{n}$. Larger n can make the mean more precise without making individual jobs less variable.

Use a separate, deliberately simple model: **prediction error = predicted runtime – actual runtime**, so negative values are possible. Assume independent normal errors with **known population SD $\sigma = 4$ seconds**. An observed sample mean of 1 second gives this approximate 95% confidence interval for the population mean error:

$$\bar{x} \pm 1.96 \frac{\sigma}{\sqrt{n}} \implies 1 \pm 1.96 \frac{4}{\sqrt{25}} = [-0.568, 2.568] \text{ s}$$

At $n = 100$, SE falls from 0.8 to 0.4 seconds. Holding the observed mean fixed at 1 for comparison gives **[0.216, 1.784] seconds**: half the width. These are assumed-model calculations, not intervals estimated from our five runtimes.

➤ More independent observations narrow uncertainty in the mean

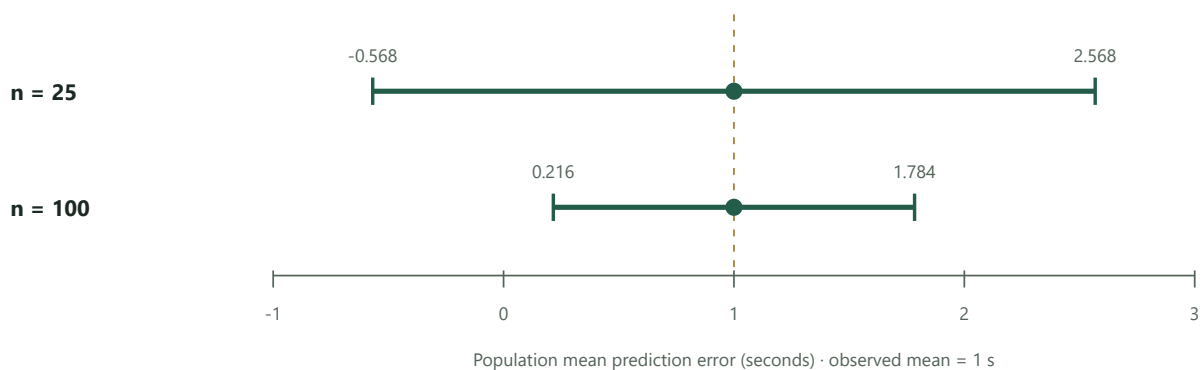


Figure 3. Known population SD = 4 seconds; assumed normal, independent prediction errors. Holding the observed mean at 1 second, quadrupling n halves interval width. These are two formula comparisons, not sampled datasets.

A **95% confidence procedure** would cover the fixed population mean in about 95% of repeated samples under its assumptions. It does not say 95% of individual errors lie in this interval, or assign a 95% posterior probability to this particular interval. A **prediction interval** concerns a future observation, a different target. Formula and coverage interpretation: NIST's confidence-interval introduction.

▼ Go deeper: unknown SD, LLN, CLT, bootstrap, and credible intervals

Population SD is usually unknown. For independent normal observations, estimating it with s leads to a **Student t interval**, with a critical value reflecting $n - 1$ degrees of freedom. Do not simply apply 1.96 to every tiny or skewed dataset. Interval methods depend on the estimator, sampling process, and assumptions.

The **law of large numbers (LLN)** explains convergence of a sample average toward its expectation under suitable conditions. The **central limit theorem (CLT)** explains why properly standardized sample means approach a normal distribution under conditions such as independent, identically distributed observations with finite variance. It does not make the original observations normally distributed; there is no universal sample size that guarantees a good approximation.

A **bootstrap** repeatedly resamples observed units with replacement and recomputes a statistic to approximate its sampling variability. Ordinary row resampling assumes appropriate independent units; grouped or time-dependent data needs suitable methods. Resampling cannot recover a missing population or automatically remove selection bias.

A Bayesian **credible interval** contains a stated amount of posterior probability for a parameter, conditional on the data, prior, and model. That interpretation differs from repeated-sampling confidence coverage. Neither type of interval automatically accounts for a wrong model or biased data.

8. Distinguish association from a cause

Covariance describes whether two numerical variables tend to vary together. Its sign indicates direction, and its units depend on both variables. **Pearson correlation** divides covariance by the product of their SDs, producing a unitless value r between -1 and 1 when both SDs are positive.

For $x = [1, 2, 3]$ and $y = [2, 4, 6]$, $r = 1$: a perfect positive linear relationship. The slope is 2, not 1. Correlation describes strength and direction of a linear association, not how many units y changes per unit of x . Zero correlation can still hide a nonlinear relationship.

Large jobs may use more memory and take longer. **Job size is a confounder**, a common influence on both. This association does not establish that assigning more memory causes slower execution. Reverse causation, selection, and measurement errors are other possibilities.

To investigate a change, define the outcome and compare like with like. A well-designed randomized A/B experiment assigns eligible jobs to configurations independently of workload, helping balance confounders in expectation. Still check sample size, interference between jobs, missing outcomes, and practical effect size. Observational comparisons require additional causal assumptions.

9. Check evidence and reproduce the arithmetic

A **hypothesis test** starts with a null hypothesis H_0 , such as “the two configurations have equal mean runtime,” and an alternative. A **test statistic** summarizes the relevant discrepancy. A **p-value** is the probability, under H_0 and the test’s assumptions, of a statistic at least as extreme as the one observed, with “extreme” defined by the test.

A p-value of 0.03 does **not** mean a 3% chance that H_0 is true. Failing to reject H_0 does not prove equality. A small p-value also does not measure effect size or business importance: a tiny speed improvement may become detectable with enough data. Report the estimated change, uncertainty, study design, and limitations together. See the [ASA’s p-value principles](#).

Choose the question and analysis before inspecting results. Trying many comparisons and showing only the smallest p-value increases false-positive risk; use appropriate multiple-testing methods and report the search. Repeatedly checking results and stopping at “significant” also needs a suitable sequential design.

▼ Go deeper: significance level, testing errors, and power

A **Type I error** rejects a true null hypothesis; a **Type II error** fails to reject a false null. These are errors of a statistical decision procedure, distinct from the monitor's job-by-job classification errors earlier.

Choose a **significance level α** before the analysis. A valid test designed at $\alpha = 0.05$ controls its probability of a Type I error at no more than 5% under the null and stated assumptions. This is a repeated-use error rate, not the probability that one rejected hypothesis was true.

Power is the probability of rejecting the null for a specified alternative: $1 - \beta$, where β is the Type II error probability for that alternative. Power depends on the effect size, variability, sample size, and test. A weakly powered study can miss a meaningful effect. Plan around a practically important change, not merely whether a p-value can cross a threshold.

Now verify the descriptive example with Python's standard library:

```
from statistics import mean, median, variance, stdev

runtimes = [2, 3, 3, 4, 8]
print(mean(runtimes), median(runtimes)) # 4 3
print(variance(runtimes))               # 5.5 (sample variance)
print(round(stdev(runtimes), 3))        # 2.345
```

`variance` and `stdev` use the sample convention; `pvariance` and `pstdev` use the population convention. The Python statistics reference documents their behavior.

Extract the ZIP, open a terminal in that folder, and run:

```
py statistics_lab.py
```

Use Python 3.10 or newer; substitute your working `python` or `python3` command if needed. No extra packages are required. The workbook prints these summaries, the Bayes examples, and the known-SD intervals. It uses expected counts, makes no random draws, trains no model, and writes no files. Its README explains assumptions and gives three small modifications to try.

10. Practice: name the denominator and the assumption

Exercise 1: change one observation

Replace 8 with 18 in [2, 3, 3, 4, 8]. Calculate the mean and median. Does their difference prove that 18 is a logging error?

▼ Reveal the calculation

The sum becomes 30, so the mean is 6 seconds; the middle value stays 3. The mean responds to the larger value. Neither this difference nor a large value alone proves an error. Inspect the job and its collection context.

Exercise 2: reverse the conditional probability

Among 10,000 jobs, 10% fail. Recall is 90% and the false-positive rate is 5%. How many true and false alerts are expected? What fraction of alerts represent failures?

▼ Reveal the denominator

There are 1,000 failed and 9,000 healthy jobs. True alerts = $0.9 \times 1,000 = 900$; false alerts = $0.05 \times 9,000 = 450$. Among 1,350 alerts, precision = $900/1,350 = 2/3$, approximately 66.67%. The denominator is all alerts, not all failed jobs.

Exercise 3: increase the sample size

In the independent, known-SD example, increase n from 25 to 100 while holding the mean and SD fixed. What happens to SE and the confidence-interval width? Does this eliminate selection bias?

▼ Reveal the reasoning

Sample size quadruples, so its square root doubles. SE falls from 0.8 to 0.4 seconds, and interval width halves. Individual-observation SD stays 4 seconds. A larger sample does not remove selection bias; the interval calculation assumes an appropriate sampling process.

11. Check your understanding

Answer all five questions, then read the explanations.

01 A monitor catches 90% of failures. Does that mean 90% of its alerts identify failures?

- A. Yes: the two probabilities are identical.
- B. No: the fraction of true alerts also depends on the base rate and false-positive rate.
- C. Yes, as long as there are enough jobs.

Answer: B. Recall conditions on actual failures. Precision conditions on alerts. Bayes' rule connects them using the base rate and the probability of false alerts.

02 Under the independent, known-SD model, what happens when sample size increases from 25 to 100?

- A. SE halves; the population SD stays the same.
- B. Both SE and population SD fall to one quarter.
- C. SE is unchanged.

Answer: A. $SE = \sigma/\sqrt{n}$. Quadrupling n doubles $\sqrt{n}$, halving SE. The variability of individual observations does not change merely because more are sampled.

03 What does 95% confidence coverage describe?

- A. 95% of observations fall inside every interval.
- B. There is a 95% posterior probability for this particular interval.
- C. About 95% of intervals from repeated samples cover the fixed parameter, under the assumptions.

Answer: C. Confidence coverage is a property of the sampling procedure. A prediction interval targets a future observation; a Bayesian credible interval has a posterior interpretation.

04 Memory use and job runtime are positively correlated. What follows?

- A. Increasing allocated memory must slow jobs.
- B. There is an association; job size could influence both.
- C. There cannot be a causal relationship.

Answer: B. Association alone does not identify a causal effect. A common influence such as job size can explain it; a causal study needs design and assumptions.

05 A test reports $p = 0.03$. Which interpretation is appropriate?

- A. The null hypothesis has a 3% chance of being true.
- B. The improvement is large and useful.
- C. Under the null and test assumptions, the probability of a statistic at least this extreme is 3%.

Answer: C. A p-value is a conditional tail probability, not a probability that the hypothesis is true, nor an effect-size measure. Read it alongside the design, effect estimate, and uncertainty.

12. Keep the question beside the number

Before trusting a statistic, identify the population, denominator, units, and assumptions. Ask whether it describes observations, estimates a population property, or supports a causal claim. Those are different jobs for evidence.

▼ Quick-reference glossary

Term	Recall cue
Population / sample	Target group / observed subset
Parameter / statistic	Population property / calculation from observations
Mean / median	Arithmetic average / middle sorted position
Variance / SD	Squared spread / spread in original units
Conditional probability	Probability within a specified condition
Prior / likelihood / posterior	Before evidence / evidence under a hypothesis / after evidence
Recall / precision	Fraction of actual positives caught / fraction of alerts that are positive
Distribution / expectation	Allocation of probability / probability-weighted average
Standard error	Sampling variability of an estimator
Confidence interval	Interval from a procedure with stated coverage under assumptions
Correlation / confounder	Linear association / common influence that can distort a causal comparison
P-value	Tail probability under the null and the test assumptions
Significance level / power	Control Type I errors / chance to detect a specified alternative

Examples, diagrams, and exercises are original and synthetic. Linked references support further study. This chapter establishes foundations; later chapters apply them to evaluation, regression, probabilistic models, and causal reasoning.

Next: Chapter 8: Calculus, optimization, and information theory connects rates of change, parameter updates, and probability losses.